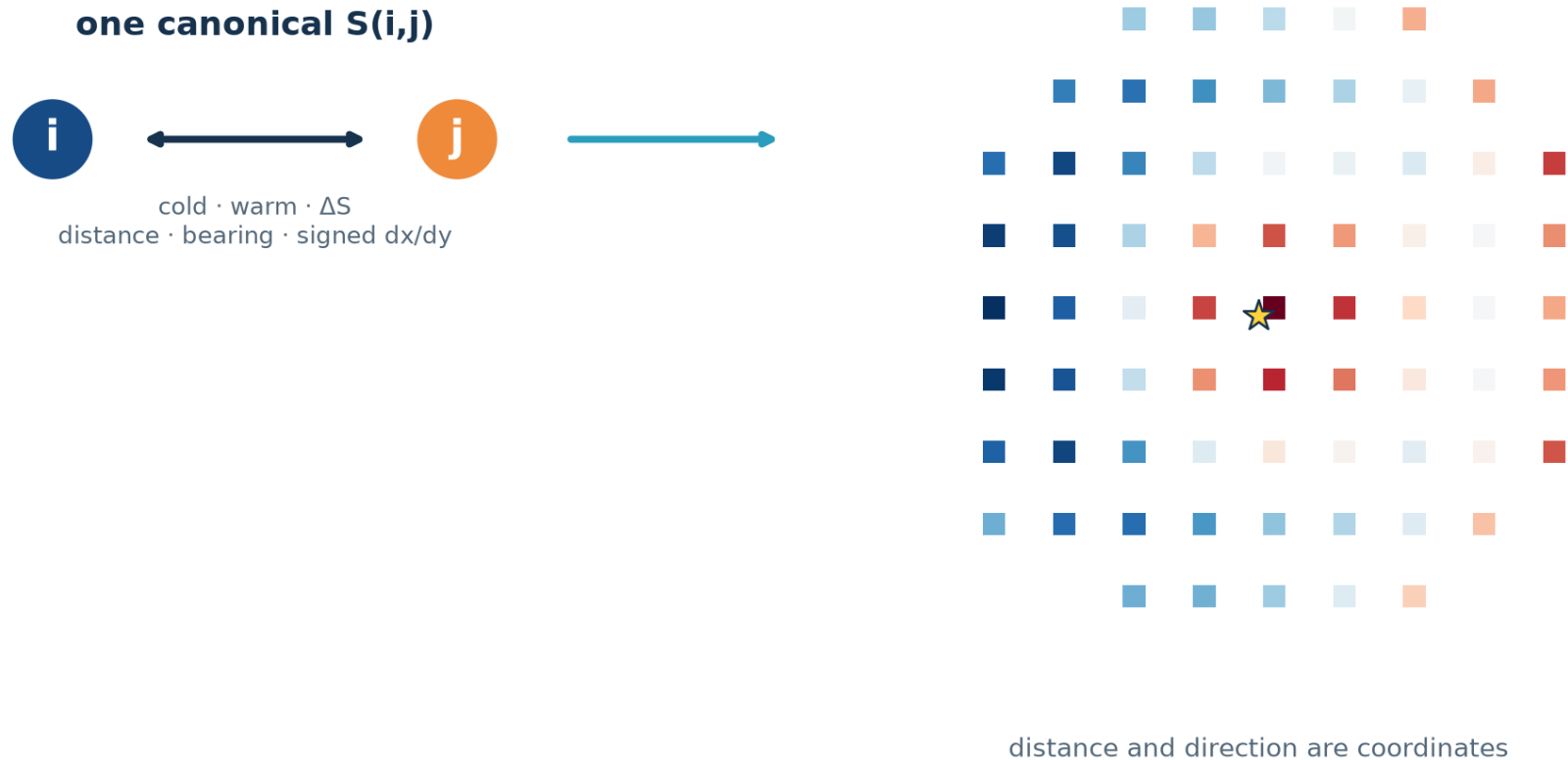


1 · Colorado local synchrony surfaces

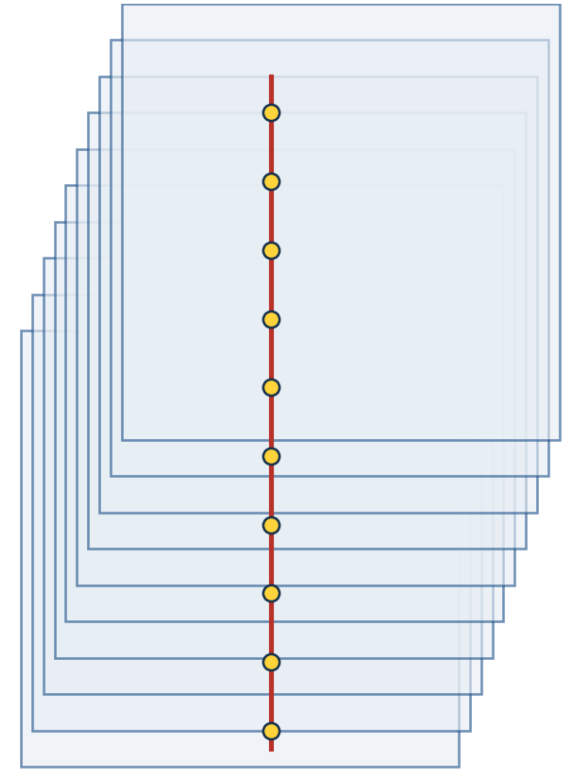
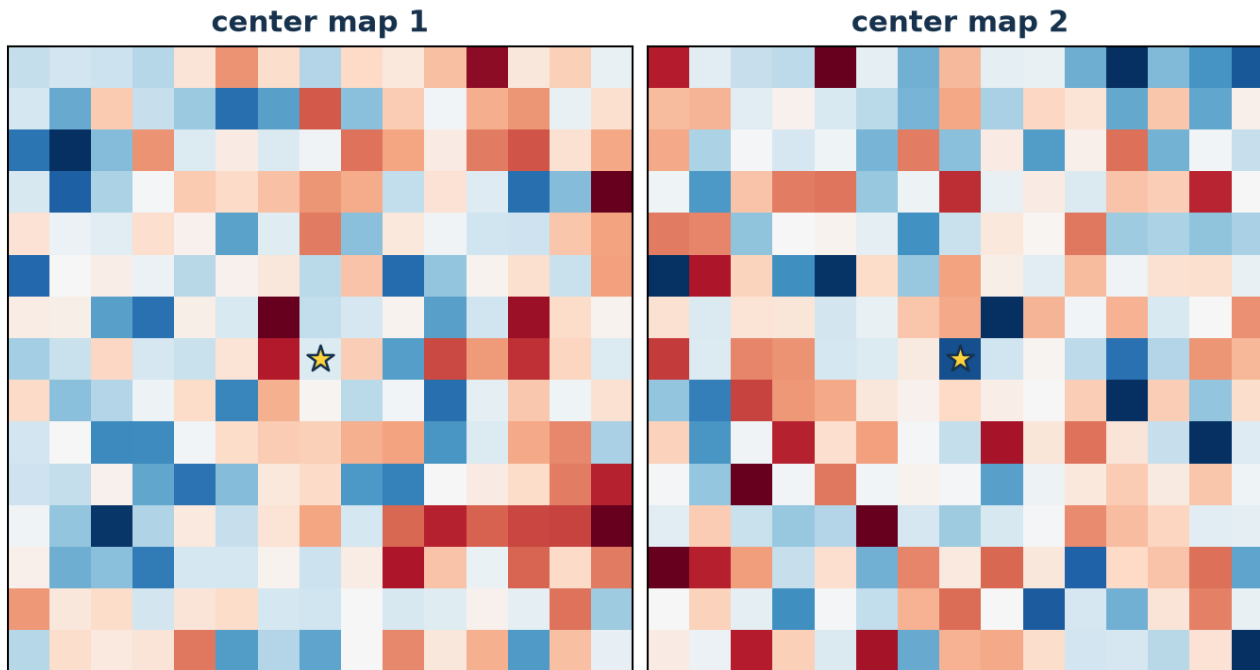
Revised Phase 2 · observed PRISM · 1 Nov 2023–30 Jan 2024



The local 2-D surface $S_p(dx, dy)$ is the scientific object. Distance and direction are coordinates. Each canonical pair contributes to both endpoint surfaces with opposite signed displacement and bearing. The 100 km limit is an observation window, not an inferred synchrony scale.

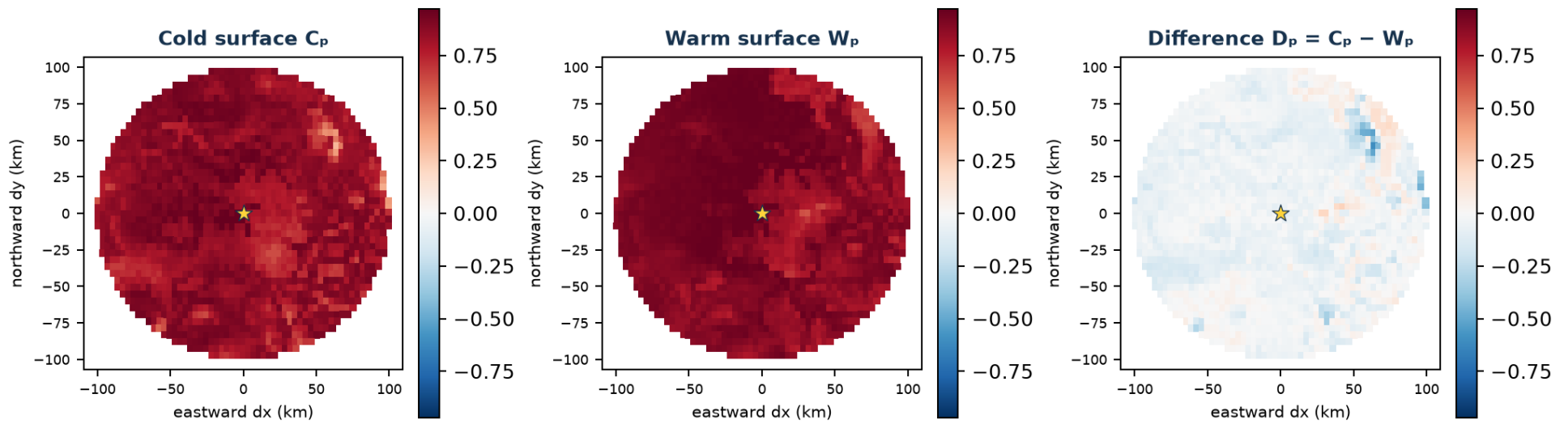
2 · Overlapping center maps generate $S_p(dx,dy)$

overlapping maps form a surface



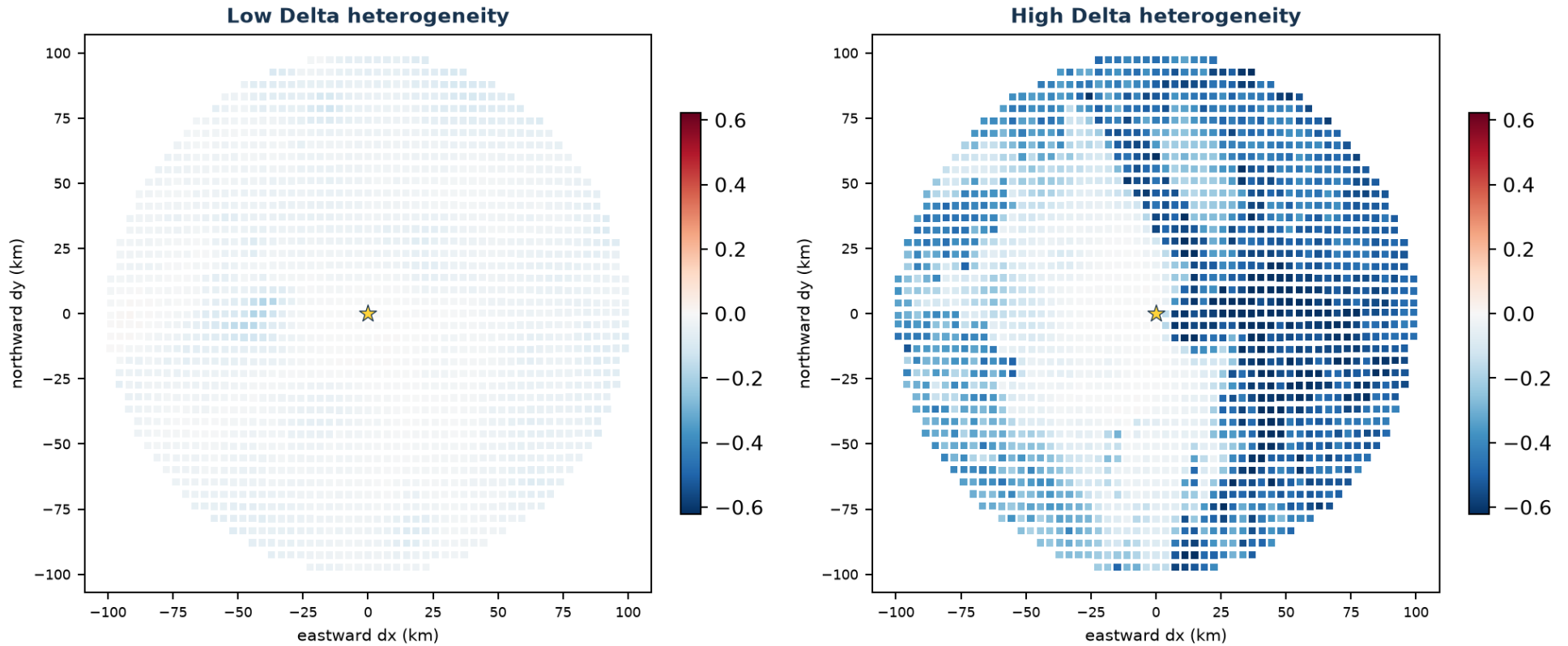
one focal value from every center map
→ organized by center displacement

3 · Real cold, warm, and Delta surfaces



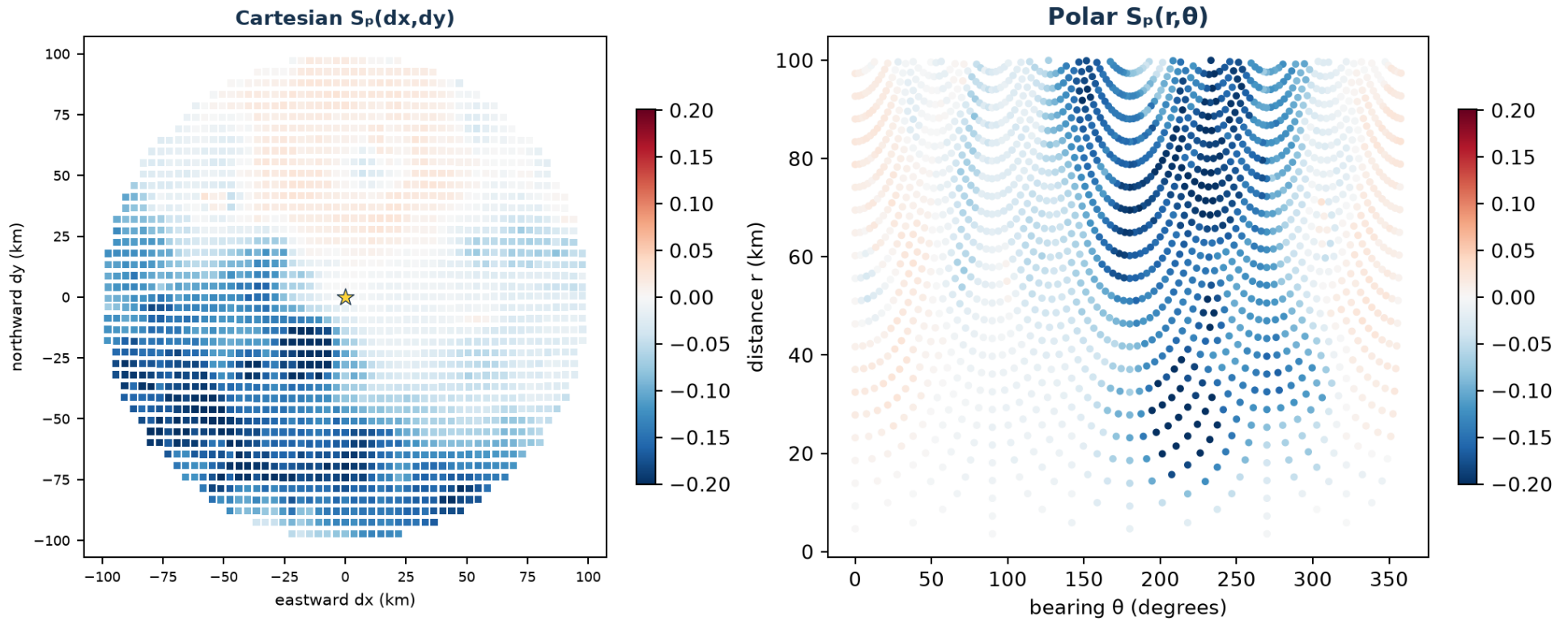
Surface analysis is never performed on Delta alone; cold and warm structure remain recoverable.

4 · Low versus high heterogeneity



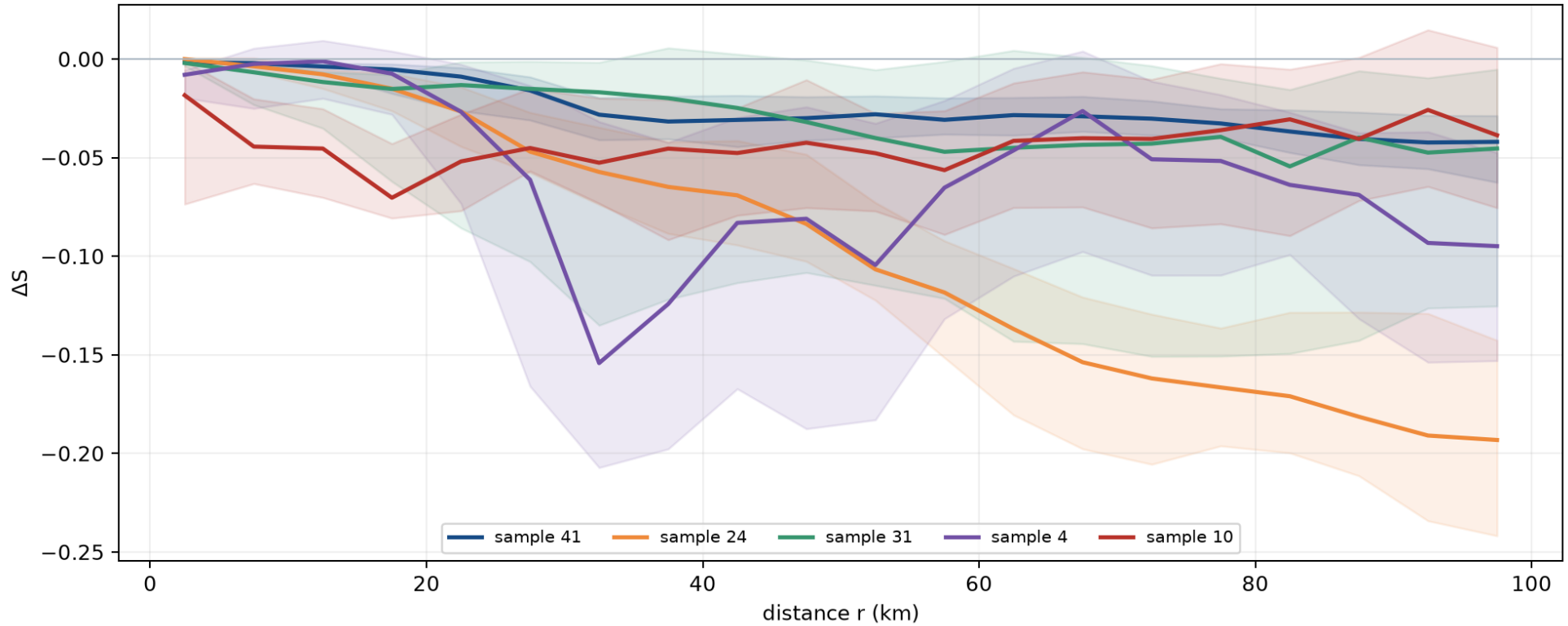
IQR and MAD describe overall dispersion but cannot show where values occur around the focal pixel.

5 · Cartesian and polar views of the same surface



6 · Continuous radial profiles

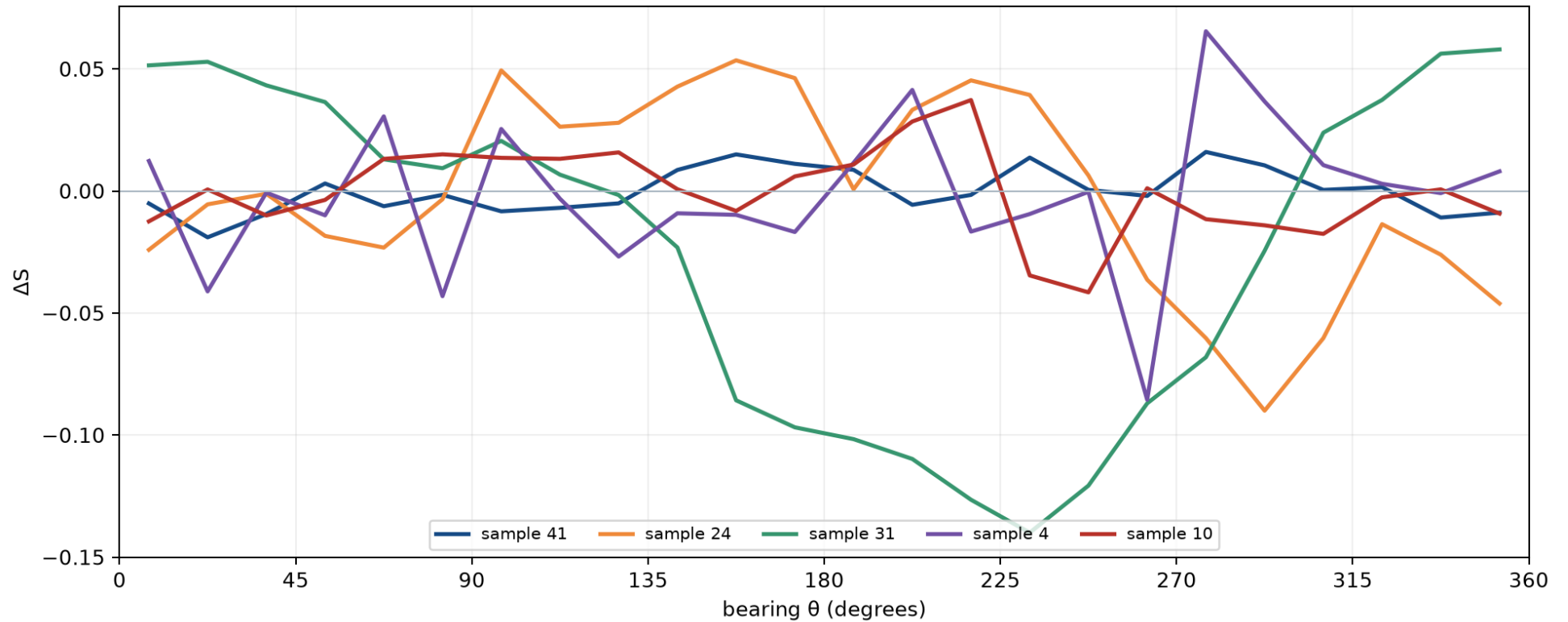
Fine radial profiles $S_p(r)$: nonmonotonicity is allowed



Five-kilometer annuli are numerical support for $S_p(r)$, not prescribed scientific scales.

7 · Direction is first-class

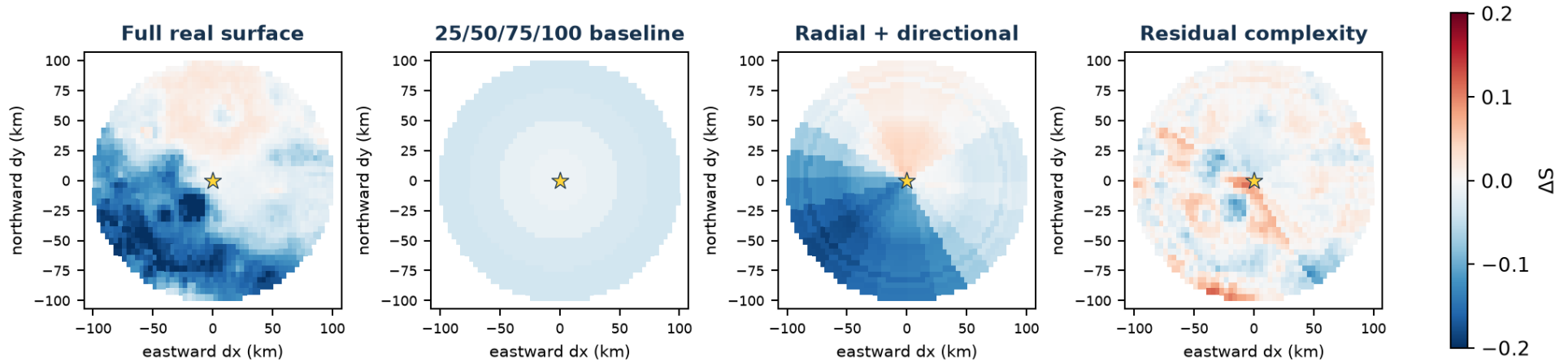
Radially detrended angular profiles $S_p(\theta)$



Angular profiles are computed after fine radial detrending so radial structure is not mistaken for direction.

8 · Why cumulative radii lose information

Worst nested-radius reconstruction in the 64-surface sample · normalized RMSE 1.09

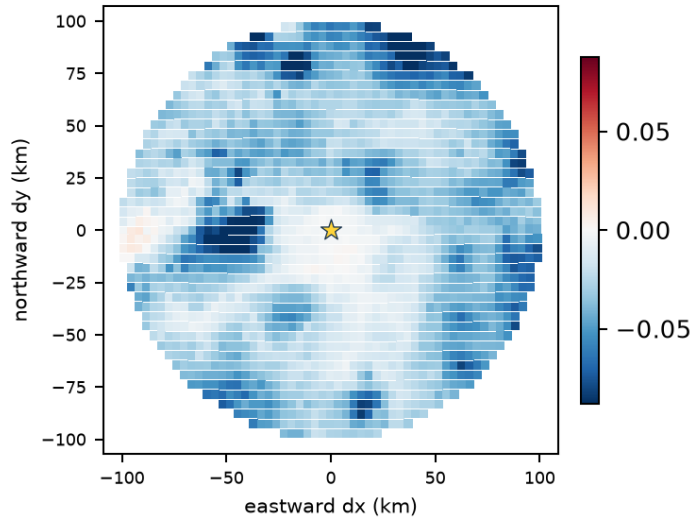


The 25/50/75/100 baseline erases one-sided and boundary-like organization even when its four values are exact.

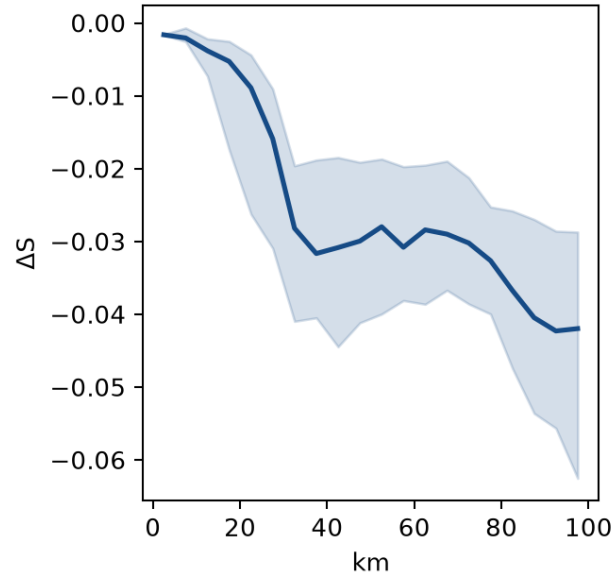
9 · Closest-to-isotropic real sample

Closest-to-isotropic sampled real surface; descriptive, not proof of isotropy

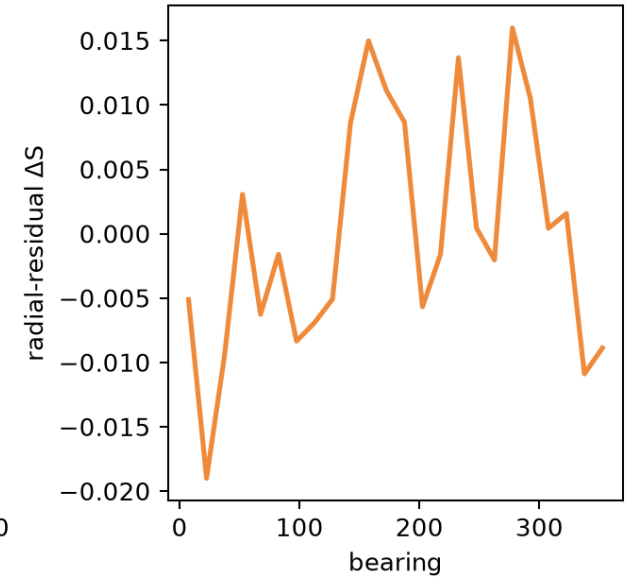
lowest-direction / low-IQR real sample



radial projection

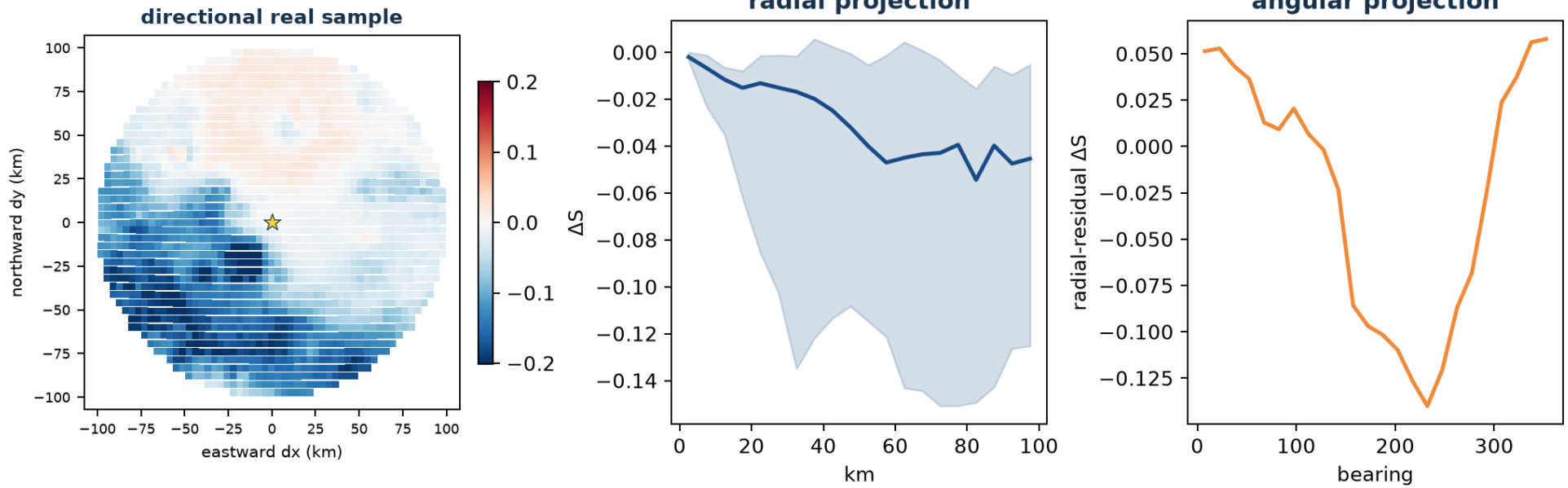


angular projection



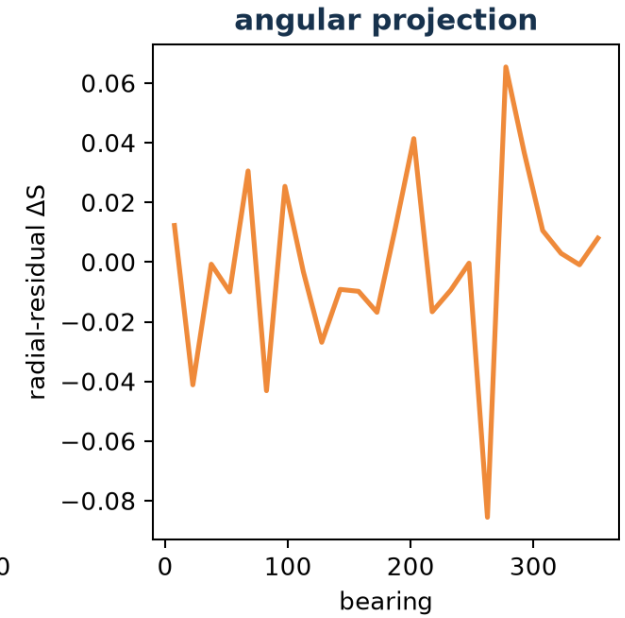
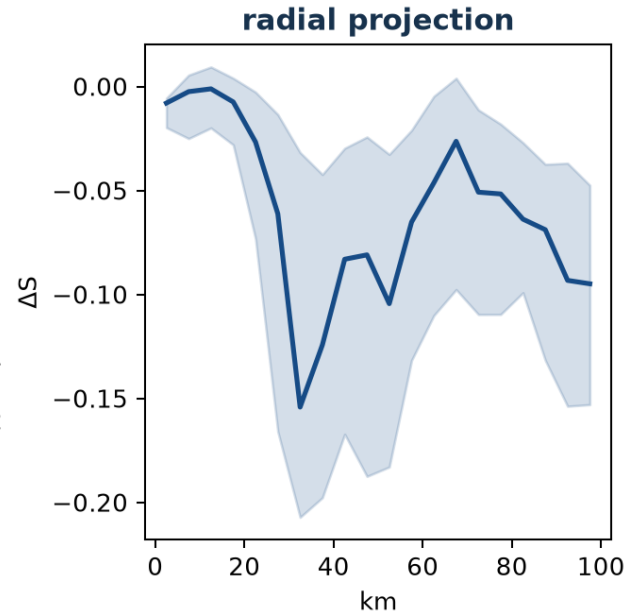
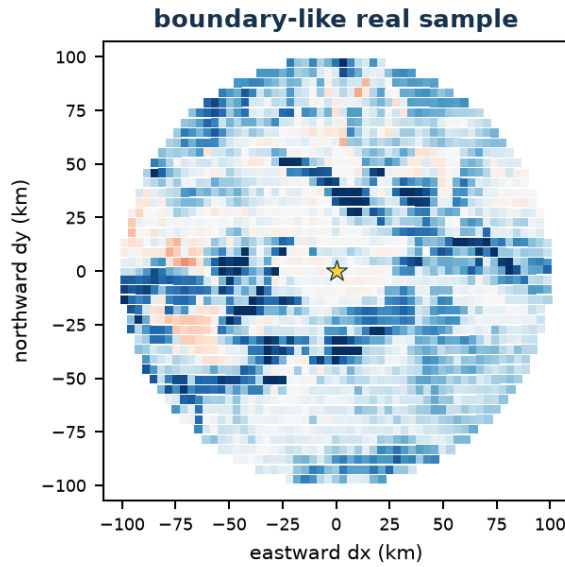
10 · Directional / anisotropic sample

Highest harmonic R^2 in the sampled real surfaces



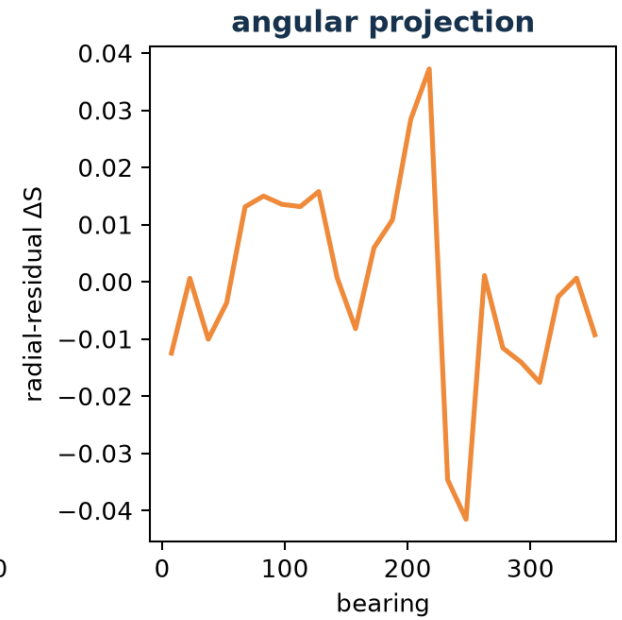
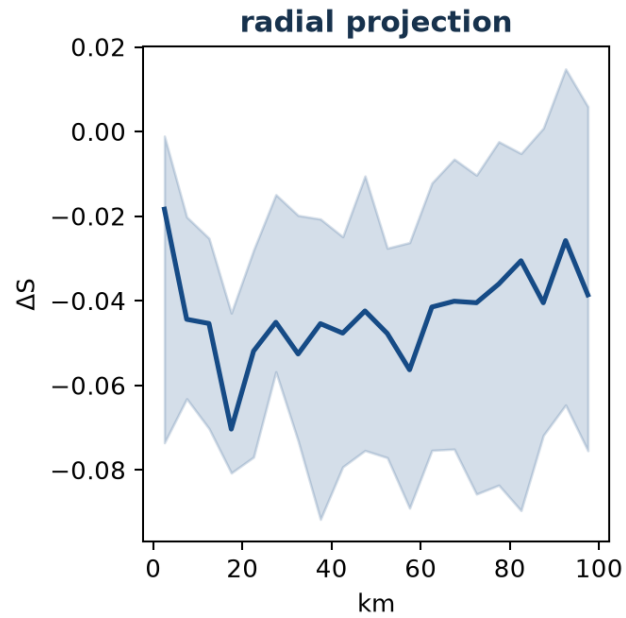
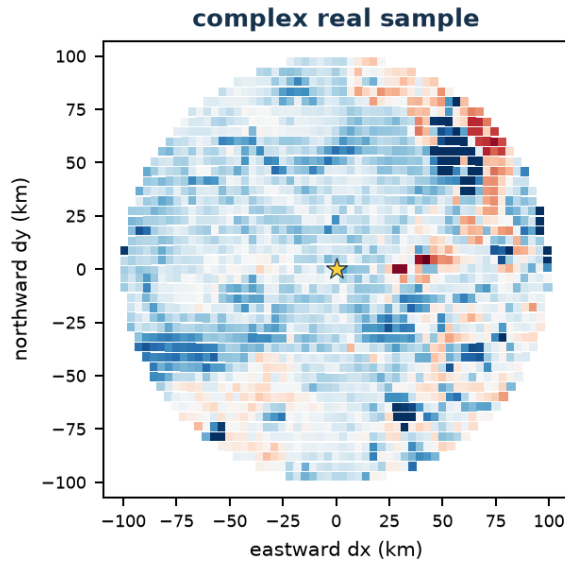
11 · Boundary-like sample

Largest half-plane contrast relative to IQR; not classified as a climate boundary



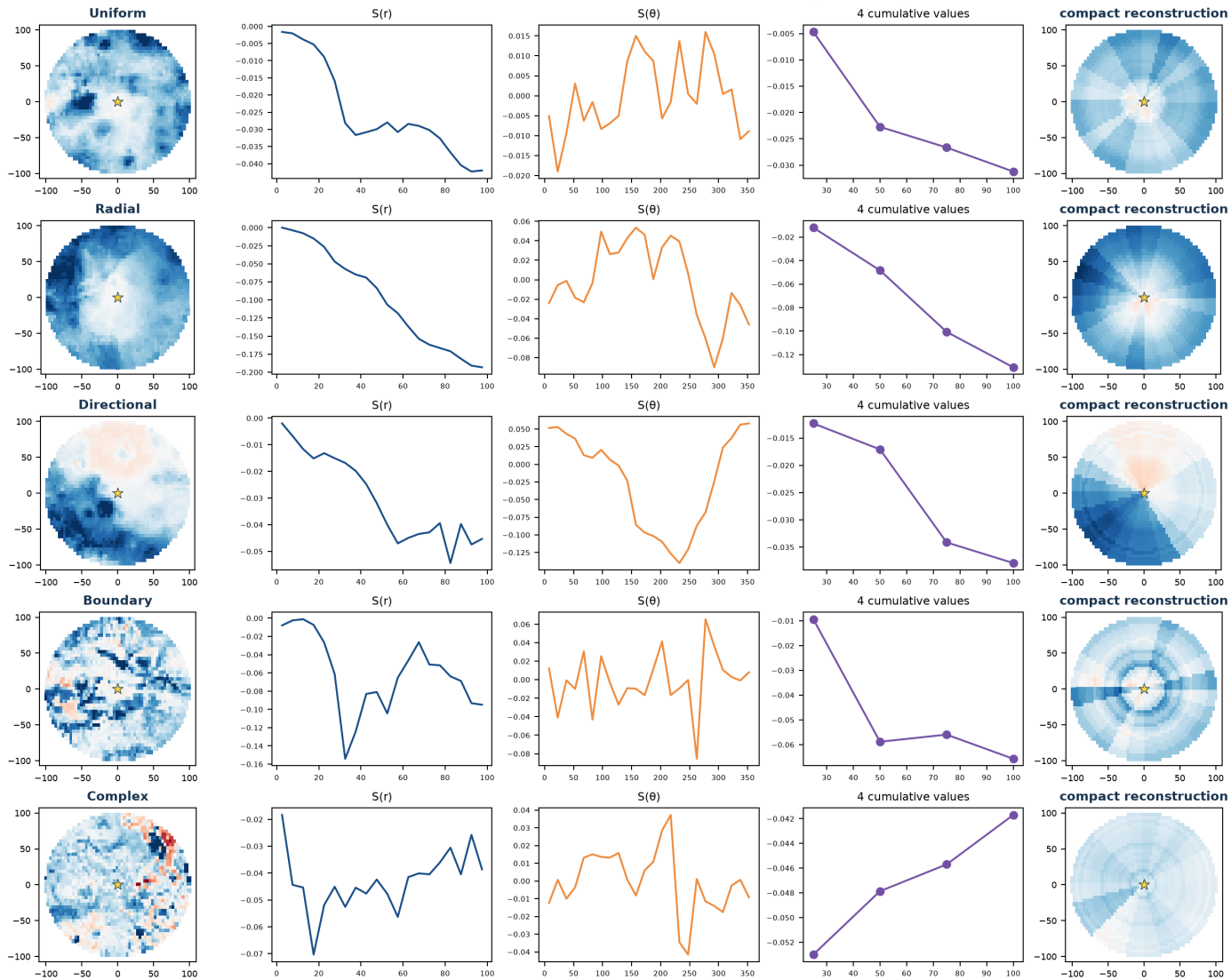
12 · Complex residual sample

Lowest low-order 2-D reconstruction R^2 in the sample

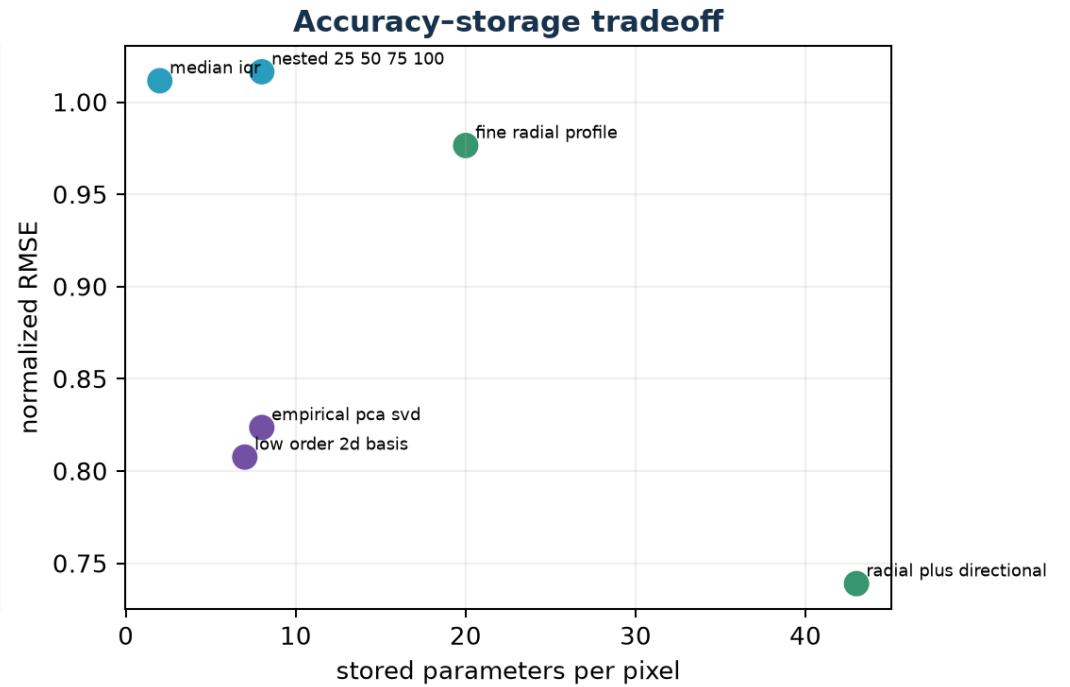
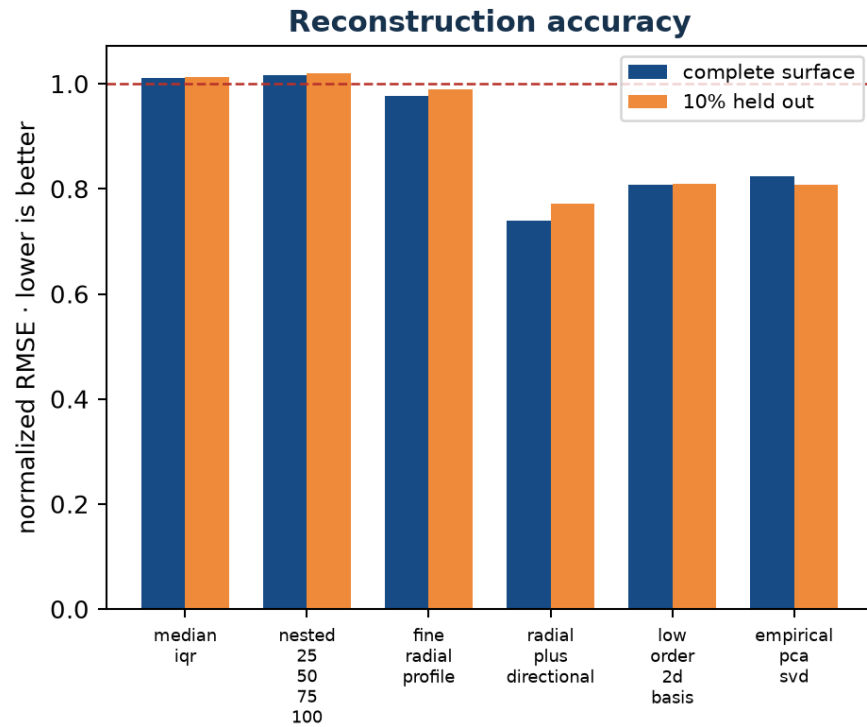


13 · Central comparison: what is lost?

What four cumulative radii lose from a 2-D local synchrony surface



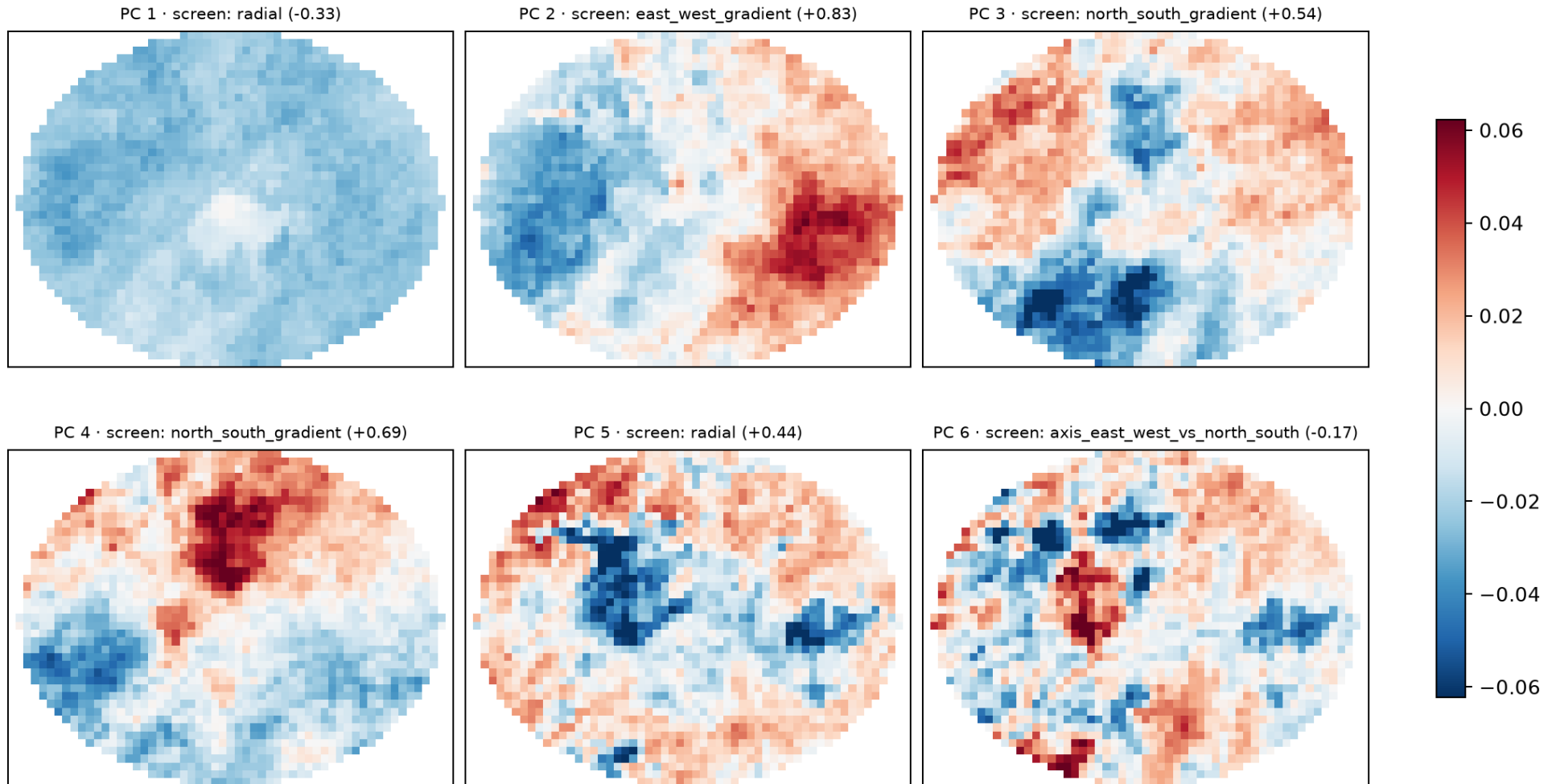
14 · Candidate compact representations



Representation	Params	RMSE	10% missing	Status
median_iqr	2	1.01	1.01	KEEP AS DIAGNOSTIC
nested_25_50_75_100	8	1.02	1.02	KEEP AS DIAGNOSTIC
fine_radial_profile	20	0.98	0.99	KEEP
radial_plus_directional	43	0.74	0.77	KEEP
low_order_2d_basis	7	0.81	0.81	EXPERIMENTAL
empirical_pca_svd	8	0.82	0.81	EXPERIMENTAL

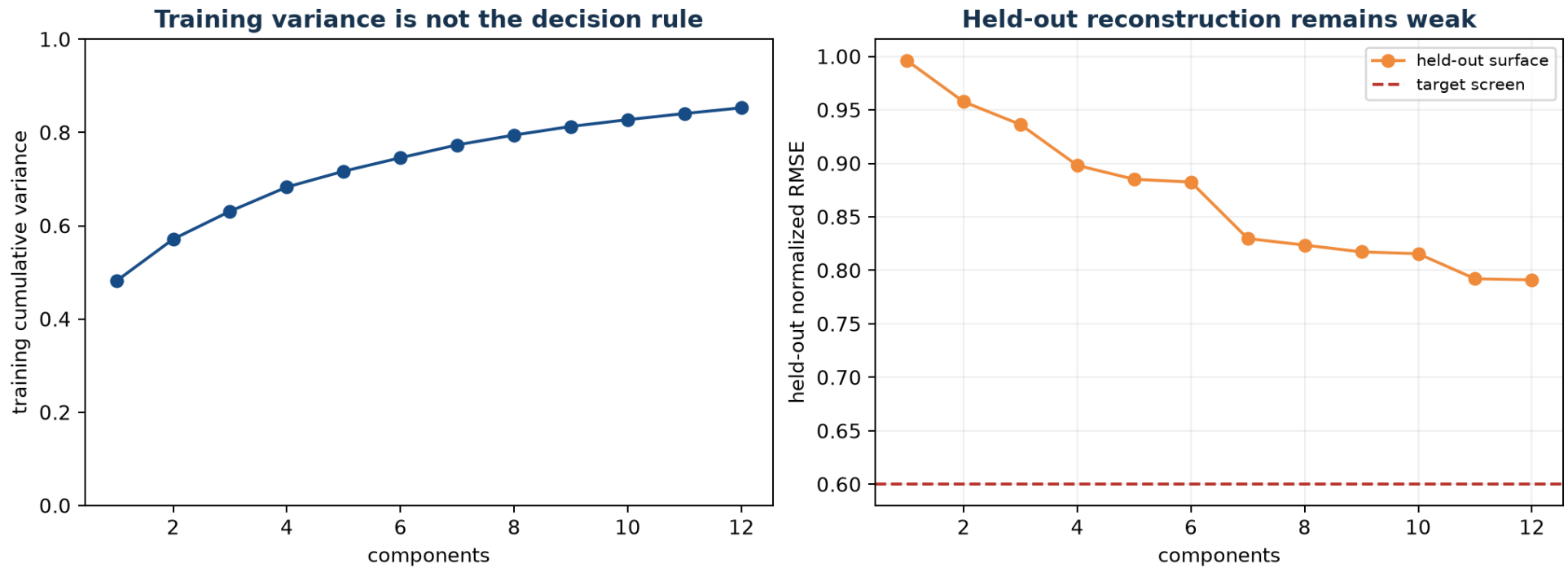
15 · Empirical PCA/SVD components

Empirical Delta basis · descriptive correlation screens, not assigned meanings



Components are inspected, not named in advance. Correlations with simple prototypes are descriptive screens only.

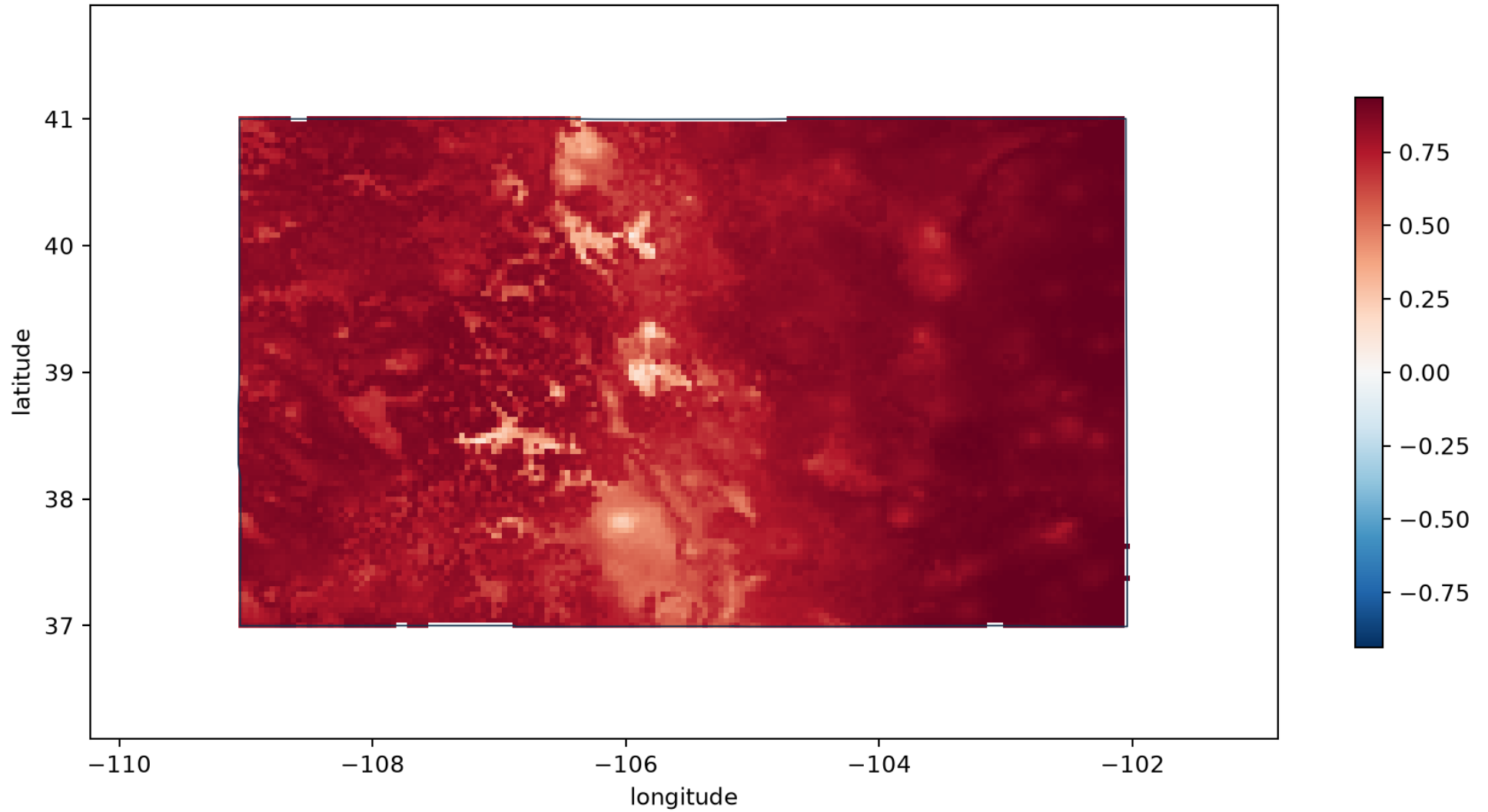
16 · Held-out reconstruction and stability



Eight components retain 79% of training variance but held-out normalized RMSE remains 0.82; split-half subspace stability is 0.31. PCA remains experimental.

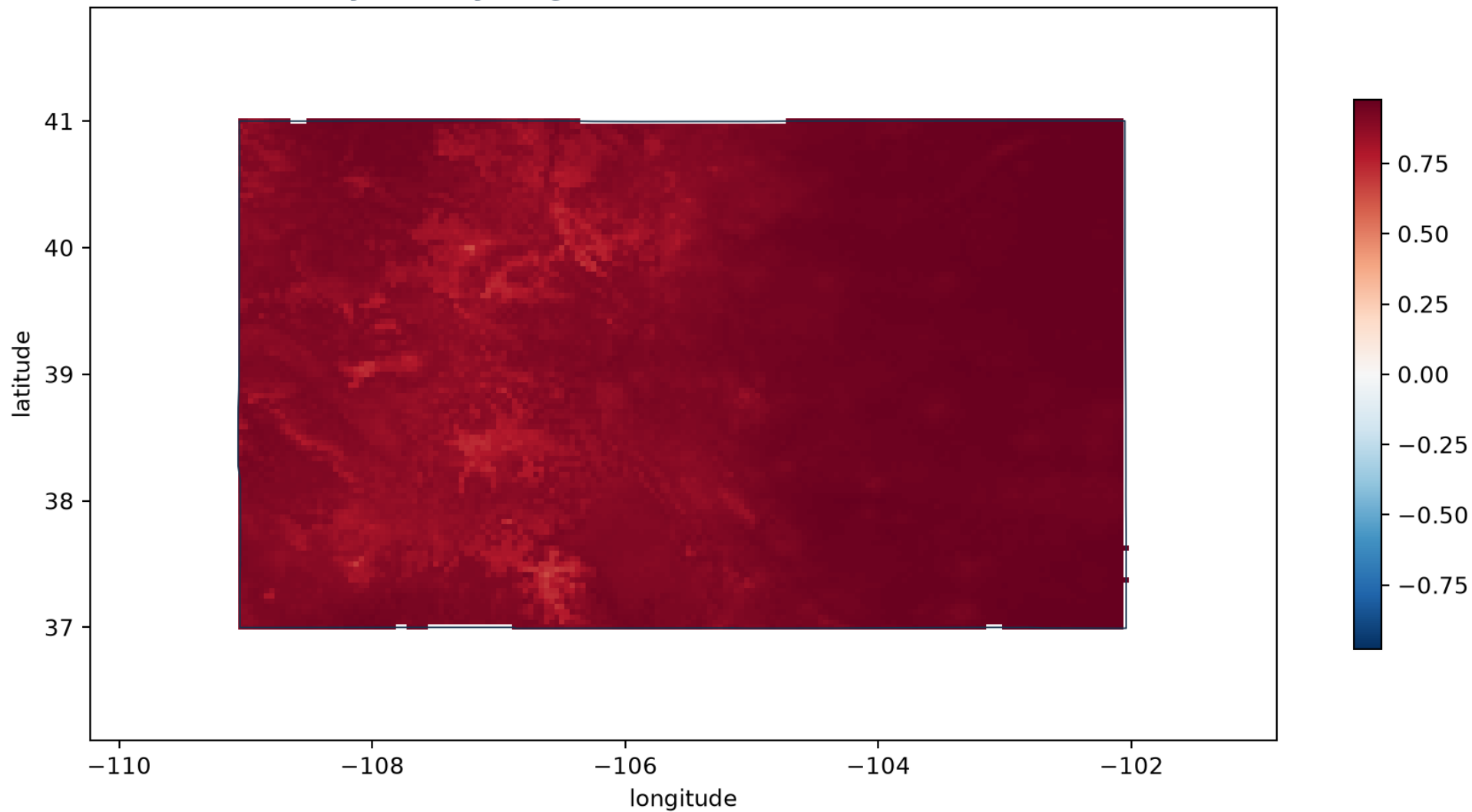
17 · Colorado cold magnitude

Cold synchrony magnitude · 100 km observation window



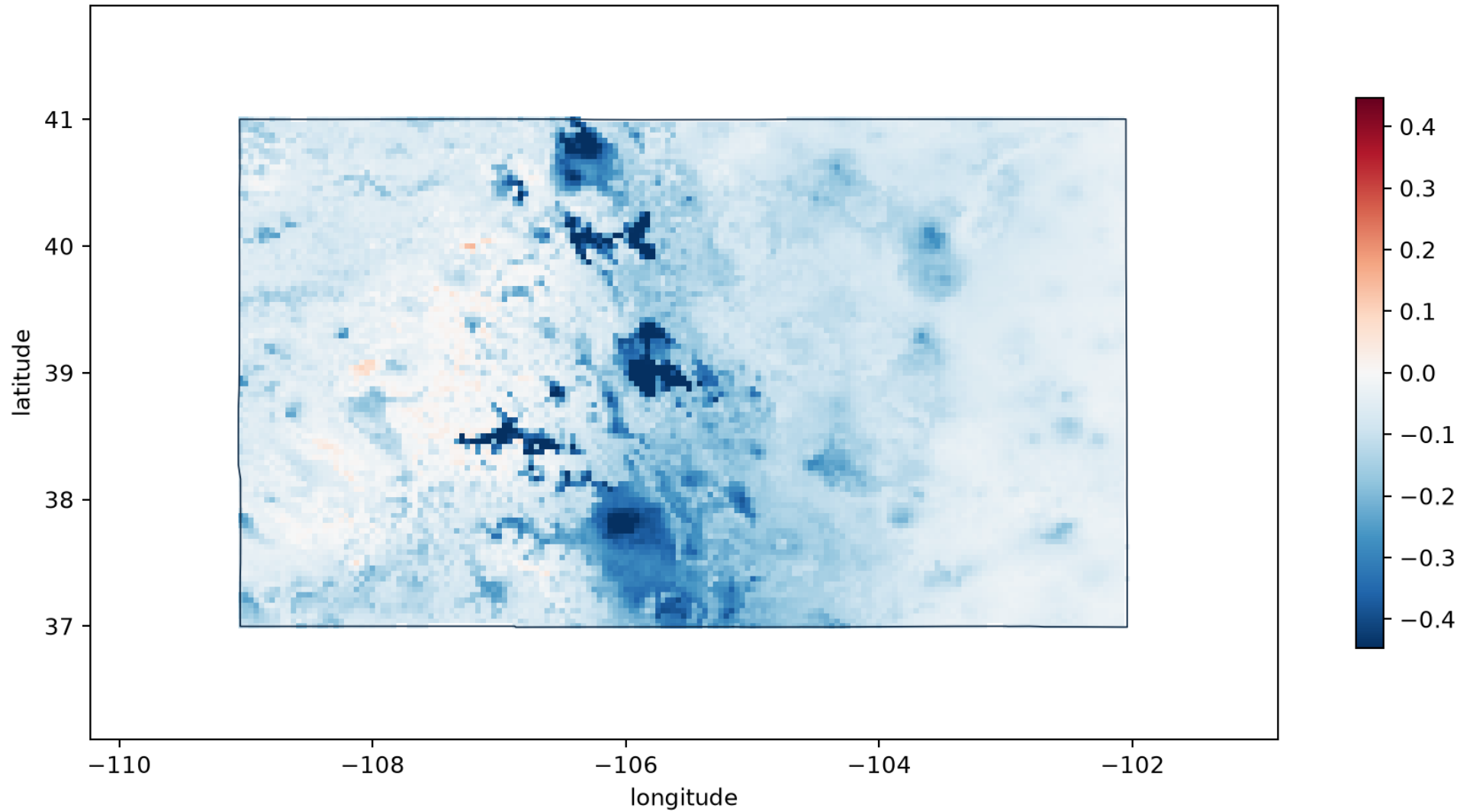
18 · Colorado warm magnitude

Warm synchrony magnitude · 100 km observation window



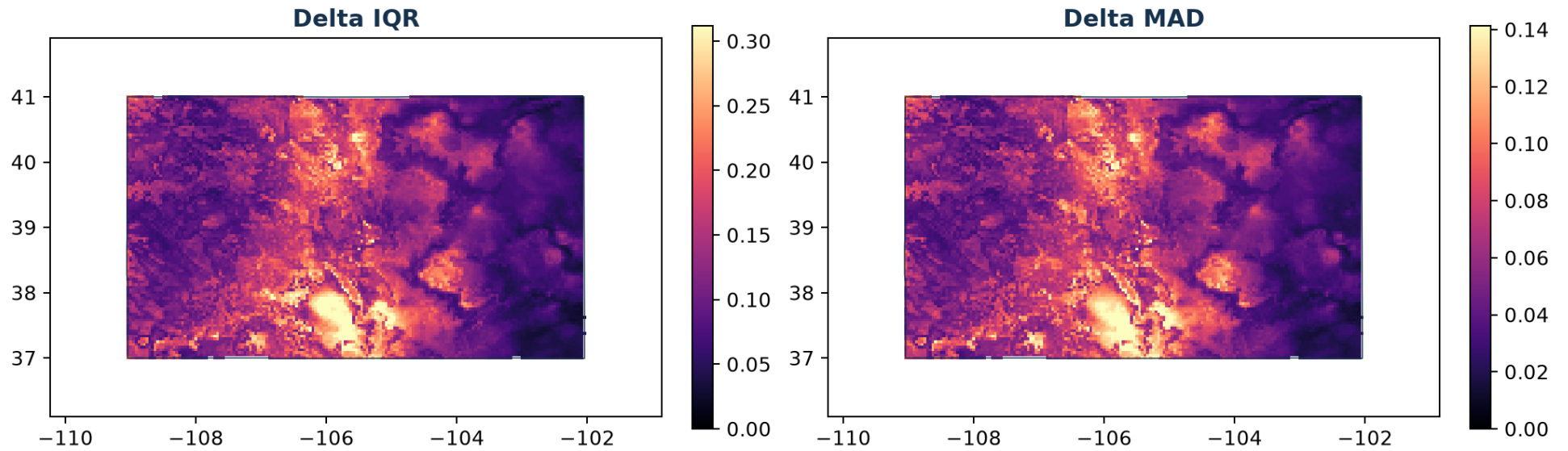
19 · Colorado Delta magnitude

Median pairwise ΔS · cold – warm



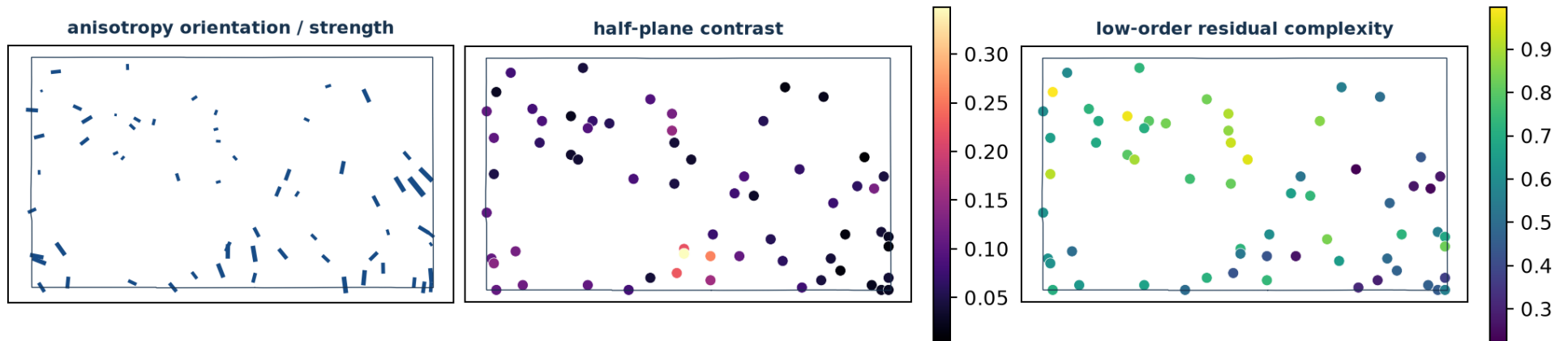
20 · Colorado overall heterogeneity

Overall heterogeneity inside the 100 km observation window



21 · Surface-shape descriptors

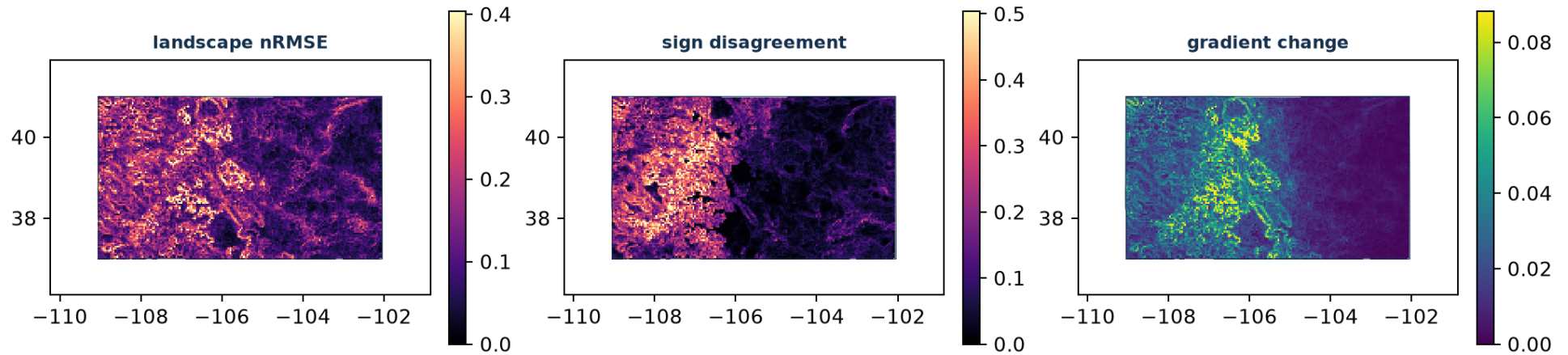
Sampled shape diagnostics · experimental, not statewide classifications



These are sparse sample diagnostics. They are not statewide regime, boundary, or characteristic-scale maps.

22 · Landscape change is separate

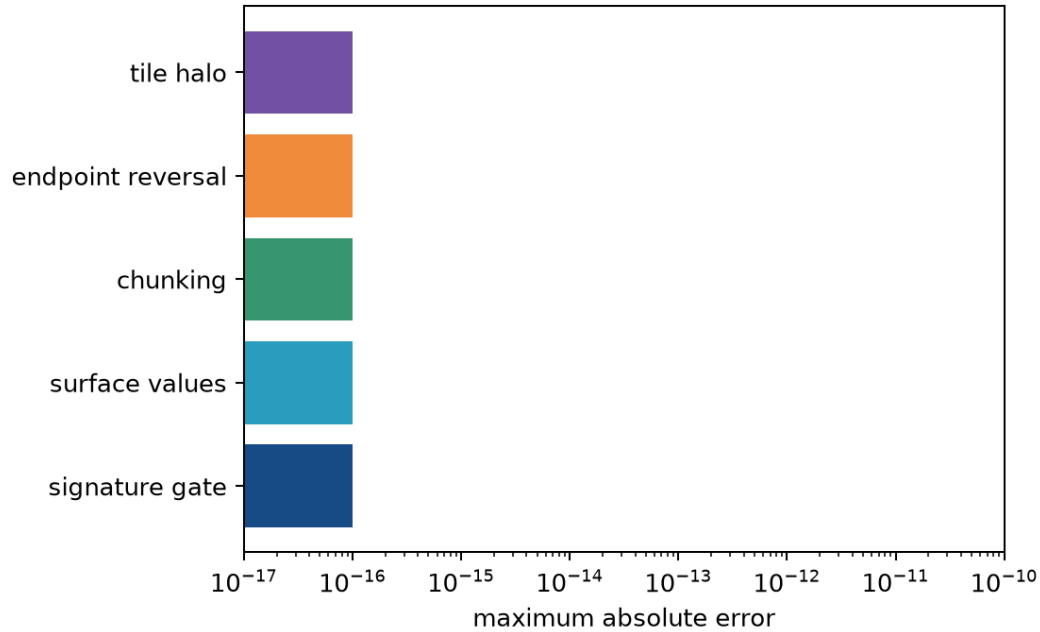
Between-center landscape change remains a separate branch



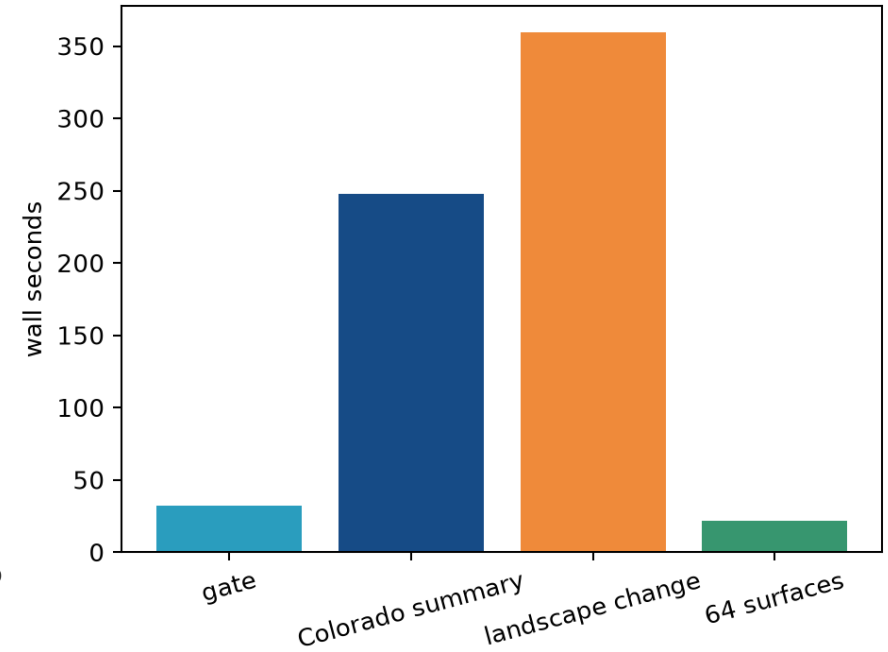
This branch compares complete center landscapes M_i and M_j ; it does not describe within-surface shape.

23 · Tile, halo, and performance validation

Real 100×100 gate



Measured execution



Run	Wall	Pairs	Peak RSS
100×100 gate	32.4 s	3,254,392	1.16 GB
Colorado summaries	248.1 s	24,922,476	1.21 GB
Landscape change	359.6 s	26,412,407	1.23 GB
64 full surfaces	12.6 s	~120k	14.8 MB stored

24 · Proposed provisional SynchronySignature

Retain canonical pair geometry during computation. Store magnitude, heterogeneity, counts/coverage, fine radial and directional projections, and residual complexity. Preserve complete cold/warm/Delta surfaces for audits and the broad sample. No final fixed basis schema is adopted.

Candidate field	Decision	Reason
magnitude + IQR/MAD	KEEP	overall level and heterogeneity
fine radial profile	KEEP	nonmonotonic, transparent projection
radially adjusted angular profile	KEEP	material directional information
anisotropy / half-plane contrast	EXPERIMENTAL	distinct and interpretable, needs temporal stability
low-order 2-D coefficients	EXPERIMENTAL	compact; residual error remains
PCA/SVD scores	EXPERIMENTAL	held-out error and split-half instability
25/50/75/100	DIAGNOSTIC	Phase 1 baseline only

25 · Phase 3 decision

Among 64 real surfaces, the experimental scale screen returned 45 candidate plateaus, 16 multiscale cases, and 3 right-censored/unresolved cases. These are diagnostics, not accepted characteristic scales.

Proceed to Colorado temporal robustness with selected seasonal windows. Test whether radial profiles, directions, half-plane contrasts, residual complexity, and any aligned basis scores persist or rotate through time. Stop before CONUS or full-history scaling.